

AIRPORT RUNWAY MANAGEMENT SIMULATOR

Week 4 Project Report
Aircraft Management System

Project	Airport Runway Management Simulator
Week	Week 4
Topic	Aircraft Management System
Technology / Concepts	Java, ArrayList, Collections Framework, Data Storage

1. Introduction

The Airport Runway Management Simulator is a Java-based software project designed to simulate basic airport runway management activities. In Week 4, the project focuses on developing the Aircraft Management System. The main objective is to create a structured system for registering aircraft, storing aircraft information, managing aircraft records, and handling landing and takeoff requests.

2. Week 4 Objectives

The main objectives of Week 4 are:

- Create an aircraft database.
- Store aircraft information.
- Register landing and takeoff requests.
- Manage aircraft records.

3. Aircraft Management System

The Aircraft Management System is a core module of the simulator. It maintains information about aircraft and provides the basic operations required to register and manage aircraft records. The module is designed so that it can later be connected with the runway management and flight management features.

4. Aircraft Database

An aircraft database is created to store the information of registered aircraft. For the initial implementation, an ArrayList can be used as an in-memory data store. Each aircraft record contains basic information needed by the simulator.

Example aircraft information includes:

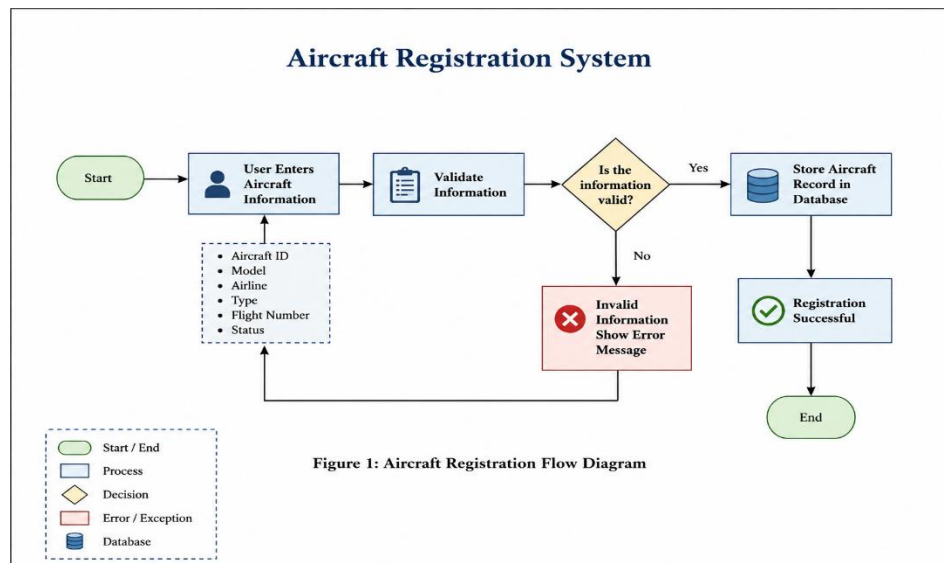
- Aircraft ID
- Aircraft Model
- Airline Name
- Aircraft Type
- Flight Number
- Current Status

Aircraft ID	Model	Airline	Type	Flight Number	Status
A001	Boeing 737	Biman	Passenger	BG101	Scheduled
A002	Airbus A320	US-Bangla	Passenger	BS201	Landing
A003	Boeing 777	Emirates	Passenger	EK585	Takeoff

Table 1: Example Aircraft Database Records

5. Aircraft Registration System

The Aircraft Registration System allows the user to enter aircraft information and register a new aircraft. After validation, the aircraft record is stored in the aircraft database. This provides a structured way to keep track of aircraft operating within the simulator.



6. Landing and Takeoff Request Registration

The system will provide a way to register landing and takeoff requests from aircraft. A request can be associated with an aircraft record and later used by the runway management system to support runway assignment and scheduling.

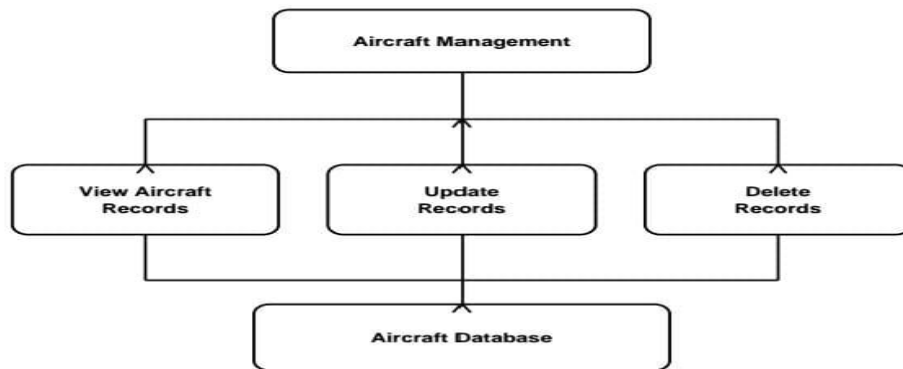
The planned request types are:

- Landing Request
- Takeoff Request.

7. Aircraft Record Management

Aircraft records need to be managed throughout the simulator's operation. The Aircraft Management System will support viewing registered aircraft and managing their records. Future versions can extend these operations to include updating and deleting records.

Aircraft Management Flowchart



Aircraft Management System

8. Technologies and Concepts to Learn

Topic	Purpose
ArrayList	Used to store and manage multiple aircraft records dynamically.
Collections Framework	Provides data structures and utilities for managing aircraft data.
Data Storage	Used to maintain aircraft information for the simulator.

9. GitHub Upload

The Week 4 implementation will be uploaded to the project GitHub repository. The required materials are:

- Aircraft Module
- Aircraft Database
- Related Java source files required to run the module.

10. Demonstration

At the end of Week 4, the project should demonstrate an Aircraft Registration System and basic Aircraft Management. The user should be able to register aircraft information, store the aircraft record, view the registered records, and demonstrate the management of aircraft data.

Demonstration Flow

Register Aircraft → Store Aircraft Information → View Aircraft Records → Manage Aircraft

11. Expected Outcome

After completing Week 4, the Airport Runway Management Simulator will have a basic Aircraft Management System. The system will be able to store aircraft information, maintain aircraft records, and register landing and takeoff requests. This module will provide the foundation for connecting aircraft data with runway and flight management features in later weeks.

12. Conclusion

Week 4 focuses on developing the Aircraft Management System of the Airport Runway Management Simulator. By using ArrayList, the Collections Framework, and data storage concepts, the project will be able to maintain aircraft records in an organized way. The completed module will support aircraft registration and management and prepare the project for further development.